

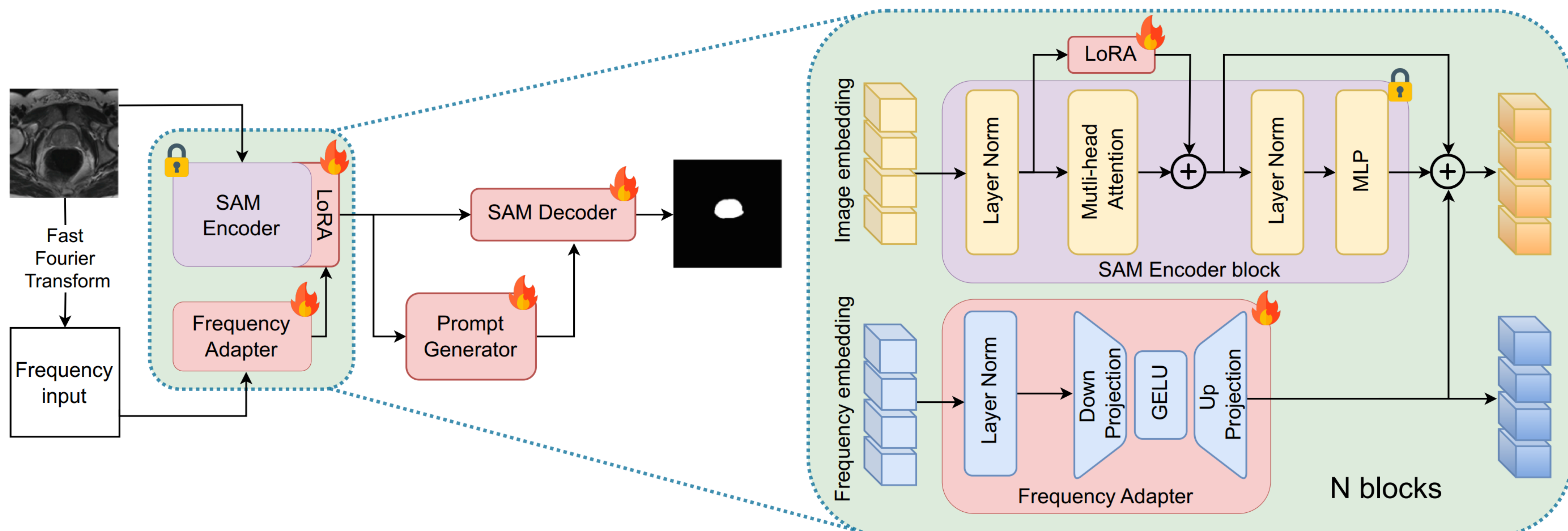
Summary

Background: Medical image segmentation models often degrade on unseen datasets because differences in scanners, imaging protocols, and patient populations introduce substantial domain shifts.

Motivation: Existing domain-generalization methods and recent SAM-based approaches mainly operate in the spatial domain, while frequency-related variations in texture, contrast, and resolution remain underexplored.

Contribution: We propose FreeSAM, combining LoRA-based SAM adaptation with a Frequency Adapter for robust cross-domain medical image segmentation.

Methodology



Overview of FreeSAM. The framework adapts SAM with LoRA and a Frequency Adapter to jointly exploit spatial and frequency-domain representations for robust medical image segmentation across unseen domains.

Performance Evaluation

Method	Venue/Year	Encoder	Messidor Base 1		Messidor Base 2		Messidor Base 3		Average		Messidor Base 1		Messidor Base 2		Messidor Base 3		Average	
			OD	OC	OD	OC	OD	OC	OD	OC	OD	OC	OD	OC	OD	OC	OD	OC
BinRushed																		
CSDG [19]	TMI'2022	UNet	93.56	81.00	94.38	83.79	93.87	83.75	93.93	82.85	89.67	75.39	87.97	76.44	89.91	81.35	89.18	77.73
ADS [23]	MICCAI'2022		94.07	79.60	94.29	81.17	93.64	81.08	94.00	80.62	90.75	77.78	90.37	79.60	90.34	79.99	90.48	79.12
MaxStyle [3]	MICCAI'2022		94.28	82.61	86.65	74.71	92.36	82.33	91.09	79.88	91.63	78.74	90.61	80.12	91.22	81.90	91.15	80.25
D-Norm [27]	CVPR'2022		94.57	81.81	93.67	79.16	94.82	83.67	94.35	81.55	92.35	79.02	91.23	80.06	92.09	79.87	91.89	79.65
SLAug [21]	AAAI'2023		95.28	83.31	95.49	81.36	95.57	84.38	95.45	83.02	93.08	80.70	92.70	80.15	92.23	80.89	92.67	80.58
CCSDG [11]	MICCAI'2023		95.73	86.13	95.73	86.28	95.45	86.77	95.64	86.57	94.78	84.94	95.16	85.68	95.00	85.98	94.98	85.53
SAMed [25]	arXiv'2023	ViT	95.28	84.24	94.11	80.21	94.84	82.60	94.74	82.35	95.41	85.26	95.36	84.25	95.38	84.76	95.38	84.76
DeSAM [8] [b]	MICCAI'2024		89.33	79.68	93.44	82.97	91.51	82.70	91.42	81.78	82.45	69.66	84.97	75.75	83.86	74.74	83.76	73.38
DeSAM [8] [p]	MICCAI'2024		91.79	80.87	92.57	82.95	93.66	84.19	92.67	82.67	81.39	67.88	83.95	76.33	79.99	73.05	84.50	72.42
DAPSAM [22]	MICCAI'2024		96.34	88.24	<u>96.10</u>	86.31	96.34	88.77	96.26	87.77	<u>96.22</u>	<u>86.74</u>	<u>96.32</u>	<u>89.59</u>	<u>96.35</u>	<u>88.12</u>	<u>96.30</u>	<u>88.15</u>
FreeSAM	–		<u>96.13</u>	<u>88.12</u>	96.29	<u>86.20</u>	<u>96.32</u>	<u>88.28</u>	<u>96.25</u>	<u>87.53</u>	96.28	87.61	96.41	90.07	96.48	88.20	96.39	88.63

(a) Segmentation performance on the multi-site RIGA+ fundus dataset.

Method	Year	Encoder	A	B	C	D	E	F	Average
In-domain	–	UNet	85.38	83.68	82.15	85.21	87.04	84.29	84.63
AdvBias [4]	MICCAI'2020	UNet	77.45	62.12	51.09	70.20	51.12	50.69	60.45
RandConv [24]	arXiv'2020		75.52	57.23	44.21	61.27	49.98	54.21	57.07
MixStyle [26]	arXiv'2021		73.04	59.29	43.00	62.17	53.12	50.03	56.78
MaxStyle [3]	MICCAI'2022		81.25	70.27	62.09	58.18	70.04	67.77	68.27
CSDG [19]	TMI'2022		80.72	68.00	59.78	72.40	68.67	70.78	70.06
SAMed [25]	arXiv'2023	ViT	80.42	<u>81.44</u>	66.75	82.09	80.19	80.17	78.51
DeSAM [8] [b]	MICCAI'2024		82.30	78.06	66.65	82.87	77.58	79.05	77.75
DeSAM [8] [p]	MICCAI'2024		82.80	80.61	64.77	83.41	80.36	<u>82.17</u>	79.02
MedSAM [18]	MICCAI'2024		72.32	73.31	61.53	64.46	68.89	61.39	66.98
DAPSAM [22]	MICCAI'2024		86.34	81.05	<u>70.81</u>	85.28	<u>82.91</u>	81.48	<u>81.31</u>
FreeSAM	–		<u>86.16</u>	82.52	74.98	86.44	83.14	83.19	82.74

(b) Segmentation performance on the multi-site prostate dataset.

Settings	Backbone	RIGA _{Bin}		RIGA _{Mag}	
		Optic disc	Optic cup	Optic disc	Optic cup
w/o FA	ViT	95.78	84.96	95.02	86.12
FA _{pha}		96.13	86.73	95.95	87.91
FA _{amp}		96.25	87.53	96.39	88.63
		Prostate			
w/o FA	ViT		79.58		
FA _{pha}			81.69		
FA _{amp}			82.74		

(c) Ablation study on frequency components.